```
clear;
clc;
close all;

%% EXPERIMENT 5: PARKS-MCCLELLAN VS FREQUENCY-SAMPLING

% Input Parameters
Fs = 10000;      % Sampling frequency (Hz)
fp = 1500;       % Passband edge frequency (Hz)
fs = 2000;       % Stopband edge frequency (Hz)
N = 40;          % Filter order

%% 1. PARKS-MCCLELLAN (EQUIRIPPLE) DESIGN
f_bands = [0, fp, fs, Fs/2] / (Fs/2);
a_bands = [1, 1, 0, 0];

b_pm = firpm(N, f_bands, a_bands);

%% 2. FREQUENCY-SAMPLING DESIGN
b_fs = fir2(N, f_bands, a_bands);

%% 3. DISPLAY NUMERICAL OUTPUTS
fprintf('=== EXPERIMENT 5: PARKS-MCCLELLAN VS FREQUENCY-SAMPLING ===\n\n');
```

```
=== EXPERIMENT 5: PARKS-MCCLELLAN VS FREQUENCY-SAMPLING ===
```

```
fprintf('Filter Order (N): %d\n', N);
```

```
Filter Order (N): 40
```

```
fprintf('Passband Edge: %d Hz\n', fp);
```

```
Passband Edge: 1500 Hz
```

```
fprintf('Stopband Edge: %d Hz\n\n', fs);
```

```
Stopband Edge: 2000 Hz
```

```
fprintf('--- PARKS-MCCLELLAN COEFFICIENTS (First 10) ---\n');
```

```
--- PARKS-MCCLELLAN COEFFICIENTS (First 10) ---
```

```
disp(b_pm(1:10));
```

```
   -0.0015    0.0072    0.0053   -0.0010   -0.0081   -0.0072    0.0037    0.0142    0.0100   -0.0090
```

```
fprintf('--- FREQUENCY-SAMPLING COEFFICIENTS (First 10) ---\n');
```

```
--- FREQUENCY-SAMPLING COEFFICIENTS (First 10) ---
```

```
disp(b_fs(1:10));
```

```
    0.0000    0.0001    0.0002   -0.0001   -0.0008   -0.0009    0.0008    0.0035    0.0029   -0.0039
```

```
%% 4. FREQUENCY RESPONSE
[H_pm, f] = freqz(b_pm, 1, 1024, Fs);
[H_fs, ~] = freqz(b_fs, 1, 1024, Fs);

%% 5. MAGNITUDE RESPONSE COMPARISON
figure('Name', 'Parks-McClellan vs Frequency-Sampling');

subplot(2,1,1);

plot(f, 20*log10(abs(H_pm) + eps), ...
    'b', 'LineWidth', 1.5);
hold on;

plot(f, 20*log10(abs(H_fs) + eps), ...
    'r--', 'LineWidth', 1.5);

grid on;
legend('Parks-McClellan (Equiripple)', 'Frequency-Sampling');

title(['FIR Filter Magnitude Response Comparison (Order N = ', ...
    num2str(N), ')']);

xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
ylim([-80 10]);
xlim([0 Fs/2]);

%% 6. PHASE RESPONSE COMPARISON
subplot(2,1,2);

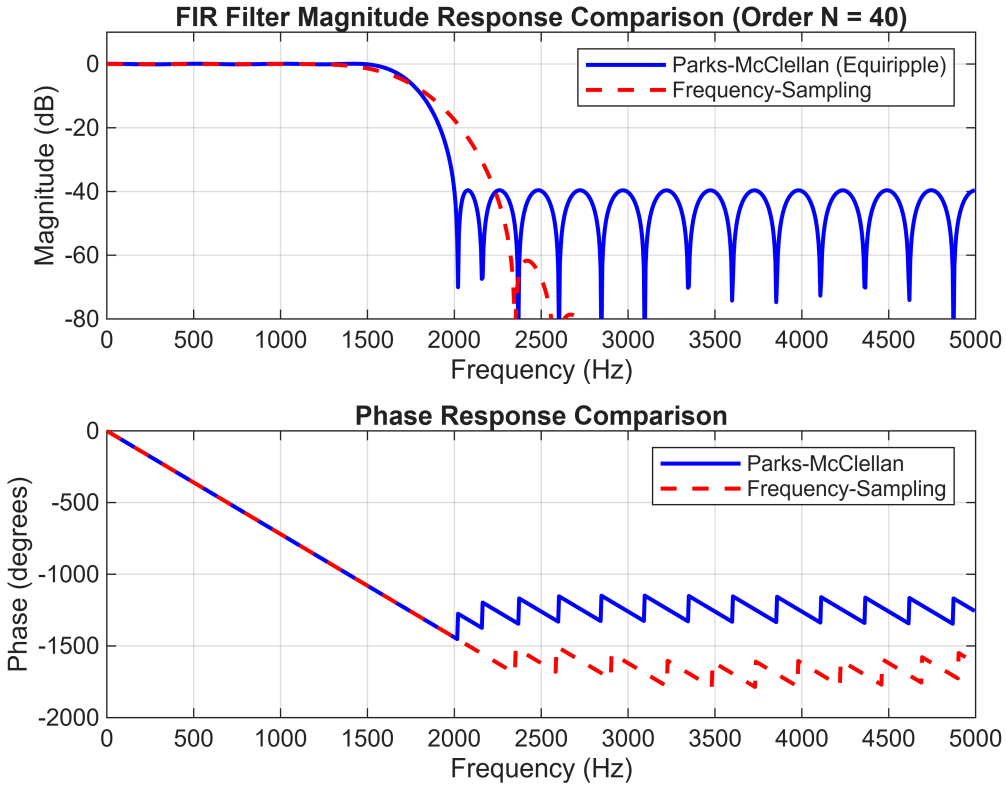
plot(f, unwrap(angle(H_pm))*180/pi, ...
    'b', 'LineWidth', 1.5);
hold on;

plot(f, unwrap(angle(H_fs))*180/pi, ...
    'r--', 'LineWidth', 1.5);

grid on;
legend('Parks-McClellan', 'Frequency-Sampling');

title('Phase Response Comparison');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
xlim([0 Fs/2]);
```

```
hold off;
```


FIR Filter Magnitude Response Comparison (Order N = 40)
Parks-McClellan (Equiripple)
Frequency-Sampling
Magnitude (dB)
0
-20
-40
-60
-80
0
500
1000
1500
2000
2500
3000
3500
4000
4500
5000
Frequency (Hz)
Phase Response Comparison
Parks-McClellan
Frequency-Sampling
Phase (degrees)
0
-500
-1000
-1500
-2000
0
500
1000
1500
2000
2500
3000
3500
4000
4500
5000
Frequency (Hz)